

Wave 5.2R Stage 4 Data-Only Residual Capacity Ladder Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

Wave 5.2R Stage 4 is complete as a valid negative result.

The campaign trained all 18 / 18 planned polished-dataset, setpoint-only, forward-direction runs without a training failure. Two hybrids appeared to beat their parameter-matched data-only controls on the trainer's scalar test surface:

- H02 improved test MAE by about 8.2% relative to C02 ;
- H08 improved test MAE by about 12.6% relative to C06 and was the

campaign's scalar winner at 0.001454936 deg . Neither candidate survived the predeclared curve-first and cancellation gates. On the canonical uniformly resampled supported-core curves, every primary hybrid failed at least one required comparison against both the frozen causal PF-A anchor and its capacity-matched direct control. No residual architecture is promoted and no stability-repeat campaign is required. The result is scientifically useful. It demonstrates that a residual network can obtain an attractive point-sampled MAE by learning a large correction that cancels the analytical anchor. That behavior is not evidence that the anchor helped the network. Stage 5 must therefore train and evaluate in the same canonical complex-coefficient and full-curve representation instead of relying on the Stage 4 point-sampled objective.

Experimental Contract

- Dataset: polished_dataset .
- Input mode: setpoints only.
- Direction: Fw only.
- Split: frozen common split signature
c1aa8718fb9bf88cc2021c121dc4f3b4010fc1d2e45ac90af5f4376aa64f8e16 .
- Curves: 675 training, 194 validation, and 97 test.
- Primary promotion surface: 96 supported-core test conditions.
- Full finiteness and decomposition population: 966 eligible forward conditions.
- Random seed: 314159 .
- Official heavy TE Curve Verification Pipeline: not executed.

The bounded closeout diagnostic reused the repository's curve-payload infrastructure but did not change any official TE Curve Verification Pipeline decision.

Causal Anchor Correction

Preparation exposed a provenance issue in the legacy Stage 3 replay. The previous artifact declared a setpoint-only contract but used measured operating averages. Stage 4 preserved that artifact for provenance and fitted a new training-only causal PF-A surface from:

- negative physical forward torque derived from the positive setpoint magnitude;
- absolute nominal input speed;
- nominal oil temperature.

The causal anchor achieves the following independent test metrics:

- raw MAE: 0.001808977 deg ;
- centered MAE: 0.001381875 deg ;
- absolute offset error: 0.000975232 deg .

The deployment envelope classifies 96 of the 97 test conditions as supported core and one as sparse/corner support. All 966 eligible forward conditions and all 27 envelope stress points produce finite outputs.

Implementation And Preflight

Stage 4 introduced

- a direct data-only pointwise control (R1);
- frozen-anchor pointwise residuals (R2);
- physically bounded pointwise residuals (R3);
- low-rank Fourier residuals (R4);
- explicit analytical-coefficient corrections (R5);
- frozen, partial-unfreeze, and full-unfreeze anchor modes;
- residual-energy losses and decomposition metrics;
- deterministic parameter-matched campaign generation;
- local, remote, preflight-only, one-batch, and enqueue-only launcher paths.

The consolidated preflight passed 16 / 16 gates. All 18 / 18 real-dataset one-batch checks passed. Zero-initialized hybrids reproduced their anchor exactly, the independent signed-torque replay error was zero, the R3 residual never exceeded 0.016873775 deg , and frozen analytical parameters were absent from the optimizer. The first campaign launch stopped before training because the central runner did not yet dispatch the new model type. The failed queue and output were preserved as diagnostic evidence. After adding and validating the dispatch entry, the clean retry completed all planned runs.

Scalar Campaign Results

ID	Family	Validation MAE [deg]	Test MAE [deg]
C01	stage4_c01_r1_compact	0.001835	0.001624
C02	stage4_c02_r1_deep	0.002001	0.001760
C03	stage4_c03_r1_compact	0.001874	0.001620
C04	stage4_c04_r1_deep	0.001828	0.001609
C05	stage4_c05_r1_compact	0.001835	0.001624
C06	stage4_c06_r1_deep	0.001915	0.001665

ID	Family	Validation MAE [deg]	Test MAE [deg]
H01	stage4_h01_r2_compact	0.002123	0.001940
H02	stage4_h02_r2_deep	0.001630	0.001617
H03	stage4_h03_r3_compact	0.058291	0.046115
H04	stage4_h04_r3_deep	0.058330	0.046188
H05	stage4_h05_r4_compact	0.002346	0.002111
H06	stage4_h06_r4_deep	0.002177	0.001965
H07	stage4_h07_r5_compact	0.001765	0.001725
H08	stage4_h08_r5_deep	0.001490	0.001455
A01	stage4_a01_r2_compact	0.002265	0.001878
A02	stage4_a02_r2_compact	0.006336	0.005241
A03	stage4_a03_r5_compact	0.002057	0.001926
A04	stage4_a04_r5_compact	0.003065	0.002846

The scalar leaderboard alone would select `h08`. That selection is rejected by the required closeout audit.

Curve-First And Cancellation Audit

The frozen causal `r0` anchor achieves, on the `96` supported-core test curves:

- raw curve MAE: `0.001824548` deg ;
- centered curve MAE: `0.001394187` deg ;
- absolute curve-mean error: `0.000981458` deg ;
- P95 curve MAE: `0.003970491` deg ;
- worst-curve MAE: `0.009475513` deg .

Hybrid	Matched control	Curve MAE [deg]	Residual/anchor RMS	Gate
H01	C01	0.060060	1.3883	fail
H02	C02	0.059691	1.3812	fail
H03	C01	0.014912	0.2687	fail
H04	C02	0.015179	0.2707	fail
H05	C03	0.060221	1.3885	fail
H06	C04	0.060441	1.3929	fail
H07	C05	0.059974	1.3871	fail
H08	C06	0.060538	1.3955	fail

All decompositions were finite on `966 / 966` curves. R3 respected its hard bound with zero violations. The other primary hybrids used residual energy larger than the analytical anchor, with population RMS ratios between `1.3812` and `1.3955`. They therefore acted as cancellation networks rather than small physics-guided corrections. The R3 candidates preserved analytical dominance, but their bounded residual could not bridge the mismatch and their raw curve MAE remained about one order of magnitude above the anchor and direct controls.

Interpretation

The apparent contradiction between scalar and curve-first results comes from the representation used by the two surfaces:

1. the analytical Polynomial-Fourier anchor is fitted and validated on one canonical uniformly resampled periodic curve per condition;
2. the Stage 4 pointwise trainer samples the longer polished curve payload;
3. an unconstrained residual can learn the difference between those representations and improve sampled-point MAE;
4. replaying that correction on the canonical curve cancels an anchor that was already accurate there. This does not invalidate the Polynomial-Fourier theory. It invalidates the tested strategy of combining that anchor with an independently point-sampled residual objective and then interpreting scalar MAE as physics-guided gain.

Exit Decision

- Stage 4 status: completed negative.
- Promoted residual architecture: none.
- Stability repeats: not required.
- Official TE Curve Verification Pipeline decision: unchanged.
- Accepted periodic GRU and periodic harmonic MLP controls: unchanged.
- Frozen causal PF-A : retained as an analytical comparator inside its supported-core envelope.

Stage 5 Entry Contract

Stage 5 is authorized as the next roadmap step, with these lessons carried forward:

1. train sine/cosine coefficient targets on the same canonical resampled curves used for evaluation;
2. reconstruct full curves inside the training objective;
3. keep offset, low-order, reducer-related, and high-order bands explicit;
4. compare direct coefficient prediction, frozen-base correction, and data-only coefficient controls at matched capacity;
5. gate every candidate on full-curve raw, centered-shape, offset, derivative, amplitude, phase, P95, and worst-case metrics;
6. reject any candidate whose learned correction dominates or cancels the analytical base without an explicit, interpretable reason.

Canonical Evidence

- Campaign output:
`output/training_campaigns/2026-07-28-10-01-40_wave52r_stage4_data_only_residual_capacity_2026_07_28/`
- Causal anchor:
`output/analysis/wave_5_2r/stage4_data_only_residual_capacity_ladder/stage4_causal_setpoint_pf_a_surface.yaml`
- Preflight summary:
`output/analysis/wave_5_2r/stage4_data_only_residual_capacity_ladder/stage4_preflight_validation_summary.json`
- Curve diagnostics:
`output/validation_checks/wave52r_stage4_data_only_residual_curve_diagnostics/2026-07-28-15-44-59__track2c_curve_payload_diagnostics/`
- Exit-gate summary:
`output/analysis/wave_5_2r/stage4_data_only_residual_capacity_ladder/closeout/stage4_exit_gate_summary.yaml`
- Detailed cancellation audit:
`doc/reports/analysis/model_development_waves/wave_5_2/physics_guided_pinn_reassessment/[2026-07-28]/stage4_data_only_residual_capacity_ladder/stage4_curve_first_and_cancellation_audit.md`

Closeout Integrity

Together, these artifacts provide the reproducible evidence for the Stage 4 negative closeout and the Stage 5 entry decision.